



Modified curved boundary scheme for two-phase lattice Boltzmann simulations

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ABSTRACT

The lattice Boltzmann (LB) method has gained much success in a variety of fields involving fluid flow and/or heat transfer. In this method, the bounce-back scheme is a popular technique for the treatment of nonslip boundaries. However, this scheme leads to a staircase representation of curved walls. Therefore many curved boundary schemes have been proposed, but mostly have detrimental effects such as mass leakage. Several correction schemes have been suggested for simulating single-phase flows, but very few discussions or studies have been made for two-phase LB simulations with curved boundaries. In this paper, the performances of three well-known schemes for dealing with curved boundaries in two-phase LB simulations are investigated through modeling a droplet resting on a circular cylinder. For all of the investigated schemes, the results show that the simulated droplet rapidly “evaporates” under the nonslip and isothermal condition, owing to the imbalance between the mass streamed out of the system by outgoing distribution functions and the mass streamed into the system by incoming distribution functions at each boundary node. Based on the numerical investigation, we formulate two modified mass-conservative curved boundary schemes for two-phase LB simulations. The accuracy of the modified curved boundary schemes and their capability of conserving mass in two-phase LB simulations are numerically demonstrated.

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1. Introduction

In the past three decades, the lattice Boltzmann (LB) method [1–5] has been developed into an efficient and powerful numerical approach for simulating fluid flow and heat transfer. Compared with conventional numerical methods, the LB method has some major advantages [6]. For example, in the LB equation the convective operator (the streaming process) is linear, while the convective terms of the Navier-Stokes equations are nonlinear. Moreover, the fluid pressure and the strain tensor are available locally in the LB method, while in conventional numerical methods the pressure field is usually obtained by solving the Poisson equation and the strain tensor is evaluated non-locally. Furthermore, the LB method is easy to parallelize because of its explicit algorithm and local interactions.

Unlike conventional numerical methods, which usually specify the macroscopic variables such as ρ and $\mathbf{u}$ at the boundaries, the LB method applies the boundary condition to the particle distribution function, giving more degrees of freedom [7] than the set of macroscopic variables. The oldest and simplest boundary scheme

in the LB community is the bounce-back scheme [8], whose concept was adopted from lattice gas models [9]. The bounce-back scheme can be classified into two strategies. In the first strategy, which is called the fullway or standard bounce-back scheme, particles are considered to travel the complete link path from a fluid node to a solid node, where the particle velocity is inverted at the collision step [7], while in the second strategy, called the halfway bounce-back scheme [8], the particles are considered to travel only half of the link distance before hitting the solid wall.

The bounce-back scheme is usually applicable to straight walls and it leads to a staircase representation of curved walls. Using a staircase approximation, the fidelity of real geometry is lost and additional errors may be introduced [10]. To improve the accuracy of LB simulations involving curved boundaries, various curved boundary schemes have been developed. The first attempt which considers the accurate shape of a curved solid wall was made by Filippova and Hanel [11]. The basic idea of their scheme is to create ghost nodes inside the solid and then apply a linear interpolation to determine the unknown distribution functions of a boundary node. However, this scheme suffers from numerical instability when the boundary node is close to the solid wall and the relaxation time is close to 1. Later, Mei et al. [12] proposed an improved curved boundary scheme based on the work of Filippova

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and Hanel, which considerably enhances the numerical stability. These two schemes have been demonstrated to be of second-order accuracy.

Subsequently, Bouzidi et al. [13] devised a curved boundary scheme that does not require an interpolation from a ghost node inside the solid. The motivation of their scheme is to interpolate the distribution functions in the interior fluid region and to include additional information about the wall location during the bounce-back process. Nevertheless, they employed separate treatments for different regions. In order to avoid discontinuity in the boundary treatment, Yu et al. [14] developed a unified scheme of Bouzidi et al.'s algorithm. These boundary treatments are known as the interpolated bounce-back scheme and maintain the second-order accuracy of the LB method. However, it introduces viscosity dependent slip at the boundary. To address this issue, Ginzburg and d'Humières [15] proposed a multi-reflection boundary scheme, which can be viewed as an enhanced interpolated bounce-back scheme. The main feature of their scheme is that it determines the coefficients of interpolation through a second-order Chapman-Enskog expansion of the interpolated distribution functions [7].

Another type of curved boundary scheme involves only a single lattice node and is free of interpolation, which is often called "one-point" or "single-node" boundary scheme. The advantage of this type of boundary scheme is its local computations. The first one-point curved boundary scheme was proposed by Junk and Yang [16]. Nevertheless, their scheme needs to compute the inverse of a matrix at each boundary node. Recently, Zhao and Yong [17] devised an alternative single-node boundary scheme. They considered a linear combination of the pre-collision and equilibrium distribution functions at the fluid boundary node and introduced five parameters, which were determined by the Maxwell iteration. Moreover, Tao et al. [18] recently developed a single-node curved boundary scheme for LB simulations of suspended particles. Some other developments of the boundary schemes for LB simulations involving curved walls can be found in a recent book written by Krüger et al. [7].

Although various curved boundary schemes have been developed, they mostly suffer from the violation of mass conservation [7], i.e., there exists mass leakage at every time step during the LB simulations. To enhance mass conservation, several improved curved boundary schemes have been proposed. Kao and Yang [19] presented an interpolation-free boundary scheme to improve the mass conservation of the boundary condition. Bao et al. [20] devised an improved version of Mei et al.'s [12] boundary scheme by redefining the density term in the equilibrium distribution function of ghost nodes in solid. In addition, Le Coupanec and Verschaevé [21] developed a mass-conservative boundary scheme for LB simulations with tangentially moving walls. Recently, Sanjeevi et al. [10] found that the interpolation-free boundary scheme proposed by Kao and Yang [19] still causes mass leakage.

The aforementioned studies are devoted to single-phase flows. In fact, since the emergence of the LB method, its application to multiphase flows has always been a very important theme. Many multiphase LB models have been proposed in the literature, such as the pseudopotential model [22–25], the color-gradient model [26–29], the free-energy model [30–33], and the phase-field model [34–37]. Among these models, the pseudopotential model is the simplest one, in which the phase separation between different phases emerges automatically as a result of particle interactions [22,23], without resorting to any technique to track or capture the phase interface. The color-gradient model employs two distribution functions to represent two different fluids and a color-gradient-based perturbation operator is used to generate surface tension as well as a recoloring step for separating different phases. The free-energy model was originally proposed by Swift et al. [30,31] based on thermodynamics considerations. In this model, the phase sep-

aration is described by a non-ideal equation of state in thermodynamic theory. The phase-field model is devised on the phase-field theory, in which the interface dynamics is described by an order parameter that obeys the Cahn-Hilliard equation or a Cahn-Hilliard-like equation.

Although some curved schemes have been utilized in the LB studies of multiphase flows [38], very few studies or discussions have been made regarding the mass conservation of multiphase LB simulations when applying the curved boundary schemes. Recently, Fakhari and Bolster [37] performed an LB study of three-phase contact line dynamics on curved boundaries using a phase-field LB model. They proposed a new algorithm to specify the three-phase contact angle on curved boundaries. As mentioned in their work, the halfway bounce-back scheme was adopted so as to eliminate mass leakage at the wall boundaries. Definitely, the halfway bounce-back scheme is capable of conserving mass in two-phase LB simulations, but it is inappropriate for certain two-phase flows, such as two-phase flows with phase change [39–41], e.g., boiling on curved surfaces [42]. The staircase approximation of the bounce-back scheme may introduce "artificial" nucleation sites and cause significant errors for boiling simulations.

The present work is therefore devoted to investigating the performances of some well-known curved boundary schemes in two-phase LB simulations. Based on the numerical investigation, two modified mass-conservative boundary schemes are formulated for two-phase LB simulations involving curved boundaries. The rest of the present paper is organized as follows. In Section 2 we introduce the LB method and three well-known types of curved boundary schemes. In Section 3 we investigate the performances of three types of curved boundary schemes in two-phase LB simulations through modeling a droplet resting on a circular cylinder. The modified mass-conservative curved boundary schemes are presented in Section 4. Numerical validation and discussion are also given there. Finally, a brief summary is given in Section 5.

2. Lattice Boltzmann method and curved boundary schemes

2.1. The lattice Boltzmann method

The LB equation using the Bhatnagar-Gross-Krook (BGK) collision operator can be written as [43]

$$f_{\alpha}(\mathbf{x} + \mathbf{e}_{\alpha}\delta t, t + \delta t) - f_{\alpha}(\mathbf{x}, t) = -\frac{1}{\tau} [f_{\alpha}(\mathbf{x}, t) - f_{\alpha}^{eq}(\mathbf{x}, t)] + \delta t F_{\alpha}(\mathbf{x}, t), \quad (1)$$

where f_{α} is the density distribution function, f_{α}^{eq} is the equilibrium density distribution function, $\mathbf{x}$ is the spatial position, $\mathbf{e}_{\alpha}$ is the discrete velocity in the α th direction, t is the time, δt is the time step, τ is the non-dimensional relaxation time, and F_{α} is the forcing term. The space is usually discretized in such a way that $\mathbf{e}_{\alpha}\delta t$ is the distance between two neighboring nodes. Therefore, after one time step $f_{\alpha}(\mathbf{x}, t)$ will arrive at its neighboring node along the lattice velocity direction $\mathbf{e}_{\alpha}$. Hence the LB equation can be split into two processes: the "collision" process

$$f_{\alpha}^*(\mathbf{x}, t) = f_{\alpha}(\mathbf{x}, t) - \frac{1}{\tau} [f_{\alpha}(\mathbf{x}, t) - f_{\alpha}^{eq}(\mathbf{x}, t)] + \delta t F_{\alpha}(\mathbf{x}, t), \quad (2)$$

and the "streaming" process

$$f_{\alpha}(\mathbf{x} + \mathbf{e}_{\alpha}\delta t, t + \delta t) = f_{\alpha}^*(\mathbf{x}, t), \quad (3)$$

where $f_{\alpha}^*(\mathbf{x}, t)$ is the post-collision state of the density distribution function.

From Eqs. (2) and (3) it can be found that the collision process is completely local and the streaming process is completely linear. Using the two-dimensional nine-velocity (D2Q9) lattice model [1–3, 43], the equilibrium density distribution function can be given

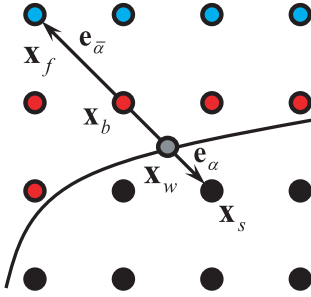


Fig. 1. Sketch of a curved wall and its neighboring lattice nodes.

by

$$f_{\alpha}^{eq} = \omega_{\alpha} \rho \left[1 + \frac{\mathbf{e}_{\alpha} \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_{\alpha} \cdot \mathbf{u})^2}{2c_s^4} - \frac{(\mathbf{u} \cdot \mathbf{u})}{2c_s^2} \right], \quad (4)$$

where $c_s = 1/\sqrt{3}$ is the lattice sound speed and the weights ω_{α} are defined as $\omega_0 = 4/9$, $\omega_{1-4} = 1/9$, and $\omega_{5-8} = 1/36$. The kinematic viscosity is related to the non-dimensional relaxation time through $\nu = c_s^2(\tau - 0.5)\delta_t$, and the macroscopic density and velocity are calculated by

$$\rho = \sum_{\alpha} f_{\alpha}, \quad \rho \mathbf{u} = \sum_{\alpha} f_{\alpha} \mathbf{e}_{\alpha} + \frac{\delta_t}{2} \mathbf{F}, \quad (5)$$

respectively, where $\mathbf{F}$ is the total force of the system and is incorporated into the LB equation through the forcing term $F_{\alpha}(\mathbf{x}, t)$ [44].

2.2. Curved boundary schemes

Fig. 1 illustrates a curved wall together with its neighboring lattice nodes. The link between a fluid boundary node $\mathbf{x}_b$ and a solid node $\mathbf{x}_s$ is cut by the wall at point $\mathbf{x}_w$. The fraction of the intersected link in the fluid region is defined as [11]

$$q = \frac{|\mathbf{x}_b - \mathbf{x}_w|}{|\mathbf{x}_b - \mathbf{x}_s|}, \quad (6)$$

where $0 < q \leq 1$ and $|\mathbf{x}_b - \mathbf{x}_s|$ depends on $\mathbf{e}_{\alpha}$.

The first LB boundary scheme for curved solid walls is proposed by Filippova and Hanel [11]. In order to determine the unknown distribution functions $f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t)$ at the fluid boundary node ($\bar{\alpha}$ denotes the opposite direction of α), Filippova and Hanel [11] defined a post-collision density distribution function at the solid node, i.e.,

$$f_{\alpha}^*(\mathbf{x}_s, t) = (1 - \chi) f_{\alpha}^*(\mathbf{x}_b, t) + \chi f_{\alpha}^{eq}(\mathbf{x}_s, t). \quad (7)$$

Then the unknown distribution functions at the boundary node can be obtained by $f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t) = f_{\alpha}^*(\mathbf{x}_s, t)$ after the streaming process. The main feature of this type of boundary scheme lies in that it introduces a fictitious equilibrium distribution function for the solid node, and Eq. (7) can be interpreted as a linear combination of the bounce-backed distribution function and the fictitious equilibrium distribution function, which is defined as

$$f_{\alpha}^{eq}(\mathbf{x}_s, t) = \omega_{\alpha} \rho(\mathbf{x}_b, t) \left[1 + \frac{\mathbf{e}_{\alpha} \cdot \mathbf{u}_s}{c_s^2} + \frac{(\mathbf{e}_{\alpha} \cdot \mathbf{u}_b)^2}{2c_s^4} - \frac{(\mathbf{u}_b \cdot \mathbf{u}_b)}{2c_s^2} \right], \quad (8)$$

where $\mathbf{u}_b = \mathbf{u}(\mathbf{x}_b, t)$ and $\mathbf{u}_s$ is a fictitious velocity. The parameter χ in Eq. (7) depends on the choice of the fictitious velocity $\mathbf{u}_s$. Filippova and Hanel [11] utilized the following choice:

$$q < 1/2: \quad \mathbf{u}_s = \mathbf{u}_b, \quad \chi = \frac{2q - 1}{\tau - 1}, \quad (9)$$

$$q \geq 1/2: \quad \mathbf{u}_s = \frac{q - 1}{q} \mathbf{u}_b + \frac{1}{q} \mathbf{u}_w, \quad \chi = \frac{2q - 1}{\tau}, \quad (10)$$

where $\mathbf{u}_w$ is the velocity at the point $\mathbf{x}_w$. In the present work, we only consider non-moving non-slip boundaries. Hence $\mathbf{u}_w = 0$. Obviously, when $q = 1/2$ ($\chi = 0$), the scheme of Filippova and Hanel reduces to the halfway bounce-back scheme [8], i.e., $f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t) = f_{\alpha}^*(\mathbf{x}_b, t)$.

It can be found that the parameter χ becomes very large when τ is close to 1 as $q < 1/2$, and then the scheme of Filippova and Hanel will suffer from severe numerical instability [45]. To solve this problem, Mei et al. [12] proposed an improved scheme, which still uses Eq. (10) for $q \geq 1/2$ but employs the following choice for $q < 1/2$:

$$\mathbf{u}_s = \mathbf{u}_f, \quad \chi = \frac{2q - 1}{\tau - 2}, \quad (11)$$

where $\mathbf{u}_f = \mathbf{u}(\mathbf{x}_f, t)$ and $\mathbf{x}_f = \mathbf{x}_b - \mathbf{e}_{\alpha} \delta_t$ is the location of the fluid node beyond the boundary node (cf. Fig. 1). Mei et al. [12] showed that the above modification significantly improves the numerical stability of the boundary scheme.

The interpolated bounce-back scheme proposed by Bouzidi et al. [13] employs an interpolation of the distribution functions in the interior fluid region. Using a linear interpolation, the boundary scheme of Bouzidi et al. is given by

$$f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t) = \begin{cases} 2q f_{\alpha}^*(\mathbf{x}_b, t) + (1 - 2q) f_{\alpha}^*(\mathbf{x}_f, t), & q < \frac{1}{2} \\ \frac{1}{2q} f_{\alpha}^*(\mathbf{x}_b, t) + \frac{2q - 1}{2q} f_{\alpha}^*(\mathbf{x}_b, t), & q \geq \frac{1}{2} \end{cases} \quad (12)$$

As $q = 1/2$, Eq. (12) reduces to the halfway bounce-back scheme. Alternatively, when a quadratic interpolation is used, the scheme takes the following form [13]:

$$f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t) = \begin{cases} q(1 + 2q) f_{\alpha}^*(\mathbf{x}_b, t) + (1 - 4q^2) f_{\alpha}^*(\mathbf{x}_f, t) \\ - q(1 - 2q) f_{\alpha}^*(\mathbf{x}_{ff}, t), & q < \frac{1}{2} \\ \frac{1}{q(2q + 1)} f_{\alpha}^*(\mathbf{x}_b, t) + \frac{2q - 1}{q} f_{\alpha}^*(\mathbf{x}_b, t) \\ - \frac{2q - 1}{2q + 1} f_{\alpha}^*(\mathbf{x}_f, t), & q \geq \frac{1}{2} \end{cases} \quad (13)$$

where $\mathbf{x}_{ff} = \mathbf{x}_b - 2\mathbf{e}_{\alpha} \delta_t$. Some other versions of the interpolated bounce-back boundary scheme can be found in the book written by Krüger et al. [7].

Another type of boundary scheme is the one-point or single-node boundary scheme. In comparison with the original single-node boundary scheme [16], Zhao and Yong [17] recently devised a simpler single-node boundary scheme. They suggested a non-convex combination of $f_{\bar{\alpha}}^*(\mathbf{x}_b, t)$ and $f_{\alpha}(\mathbf{x}_b, t)$ at the fluid boundary node, which is defined as follows:

$$f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t) = \frac{2q}{1 + 2q} f_{\bar{\alpha}}^*(\mathbf{x}_b, t) + \frac{1}{1 + 2q} f_{\alpha}(\mathbf{x}_b, t). \quad (14)$$

In this scheme, the parameters/coefficients are determined based on an analysis using the Maxwell iteration. Obviously, Eq. (14) only involves local computations. In addition, it can be found that Eq. (14) does not reduce to the halfway bounce-back scheme when $q = 1/2$.

The scheme of Mei et al. [12], the scheme of Bouzidi et al. [13], and the scheme proposed by Zhao and Yong [17] can be regarded, respectively, as the representative schemes of the aforementioned three types of boundary schemes. For convenience, these boundary schemes are referred to as the MLS scheme, the Bouzidi scheme, and the Zhao-Yong scheme, respectively. In addition, we use “L-Bouzidi” and “Q-Bouzidi” to represent the Bouzidi scheme that utilizes a linear interpolation and a quadratic interpolation, respectively.

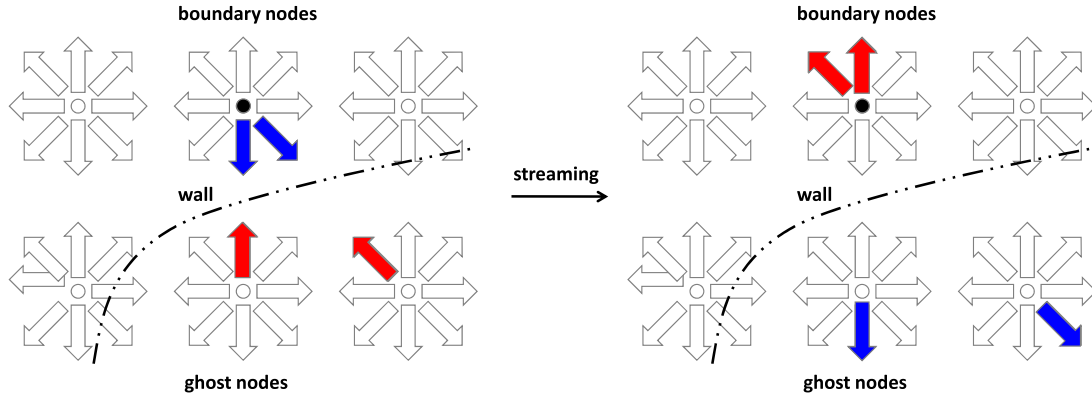


Fig. 2. Sketch of the distribution functions that propagate across a curved wall through a boundary node. The blue arrows represent the outgoing distribution functions, which propagate towards the ghost nodes in solid, while the red arrows represent the incoming distribution functions that propagate from the ghost nodes to the boundary node. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

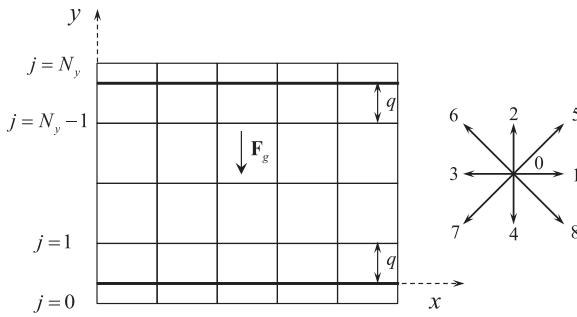


Fig. 3. Configuration of static flow under gravity with an arbitrary q .

3. Numerical investigation

3.1. Formulation of mass leakage

Fig. 2 illustrates the distribution functions that will propagate across a curved wall through a boundary node (denoted by a solid black circle) during the streaming step. In the figure the blue arrows represent the outgoing distribution functions, which propagate from the boundary node to the ghost nodes in solid, while the red arrows represent the incoming distribution functions that propagate from the ghost nodes to the boundary node.

According to Fig. 2, we can formulate the mass leakage at the boundary node as follows:

$$\Delta m(\mathbf{x}_b, t + \delta_t) = \sum_{\text{outgoing}} f_{\alpha}^*(\mathbf{x}_b, t) - \sum_{\text{incoming}} f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t), \quad (15)$$

in which the first term on the right-hand side of Eq. (15) is the amount of mass streamed out of the system by the post-collision outgoing distribution functions, while the second term on the right-hand side of Eq. (15) is the amount of mass streamed into the system by the incoming distribution functions. For the halfway bounce-back scheme [8], which gives $f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t) = f_{\alpha}^*(\mathbf{x}_b, t)$, the mass leakage is zero. However, as previously mentioned, the halfway bounce-back scheme may cause considerable numerical errors for certain two-phase flows. For example, the staircase approximation of the bounce-back scheme may introduce “artificial” nucleation sites for boiling or condensation on curved surfaces.

To illustrate the mass leakage caused by different boundary schemes, we firstly consider a test of single-phase flows, i.e., a static flow under gravity, which is illustrated in Fig. 3. The grid system is taken as $N_x \times N_y = 100 \times 50$ and the non-dimensional relaxation time τ is set to 1. The aforementioned three types of bound-

ary schemes are employed at the upper and lower solid walls and the periodic boundary condition is applied in the x direction. The gravity force is given by $\mathbf{F}_g = (0, -\rho g)$, in which the gravitational acceleration is chosen as $g = 0.001$.

In our simulations, the fraction q defined by Eq. (6) is taken as $q = n/10$ with $n = 1, 2, \dots, 9$. The variations of the normalized mass of the system are plotted in Fig. 4(a), (b), (c), and (d) for the MLS scheme, the L-Bouzidi scheme, the Q-Bouzidi scheme, and the Zhao-Yong scheme, respectively. As can be seen in these figures, the investigated boundary schemes exhibit different performances when different values of q are applied. More specifically, for the MLS scheme, the L-Bouzidi scheme, and the Q-Bouzidi scheme, the mass of the system is conserved in the case of $q = 0.5$, but decreases in other cases, which is attributed to the fact that the MLS scheme and the Bouzidi scheme (regardless of using the linear or quadratic interpolation) reduce to the halfway bounce-back scheme in the case of $q = 0.5$.

To quantify the numerical results, an averaged mass change rate is evaluated at $t = 60000\delta_t$, which is defined as $\varepsilon = |m - m_0|/m_0 \times 100\%$. Here m is the mass of the system at $t = 60000\delta_t$. The results are displayed in Fig. 5. From the figure we can clearly see that the mass change rate caused by the Zhao-Yong scheme is much larger than those produced by other schemes. Such a finding indicates the single-node type of boundary schemes may perform worse than other types of curved boundary schemes in terms of mass conservation although it has the advantage of local computations. Moreover, Fig. 5 also shows that there are no significant differences between the results of the MLS and Bouzidi schemes when $q < 0.5$, but the MLS scheme performs better as $q > 0.5$. In other words, the overall mass loss caused by the MLS scheme will be smaller than that caused by the Bouzidi scheme when the fraction q in the LB simulations varies from 0 to 1.

3.2. Simulations of droplet resting on a circular cylinder

In this section, numerical simulations are carried out to investigate the performances of the MLS scheme, the Bouzidi scheme, and the Zhao-Yong scheme in two-phase LB simulations. The test of a droplet resting on a circular cylinder is employed. Under the non-slip and isothermal condition, no evaporation will occur when a droplet is placed on a circular cylinder. However, if there exists mass leakage such as mass loss in the LB simulations, the droplet will gradually become smaller, like an evaporating droplet. In our simulations, the isothermal pseudopotential multiphase LB method [3,6,22,23,25,46] is adopted, in which the fluid-fluid inter-

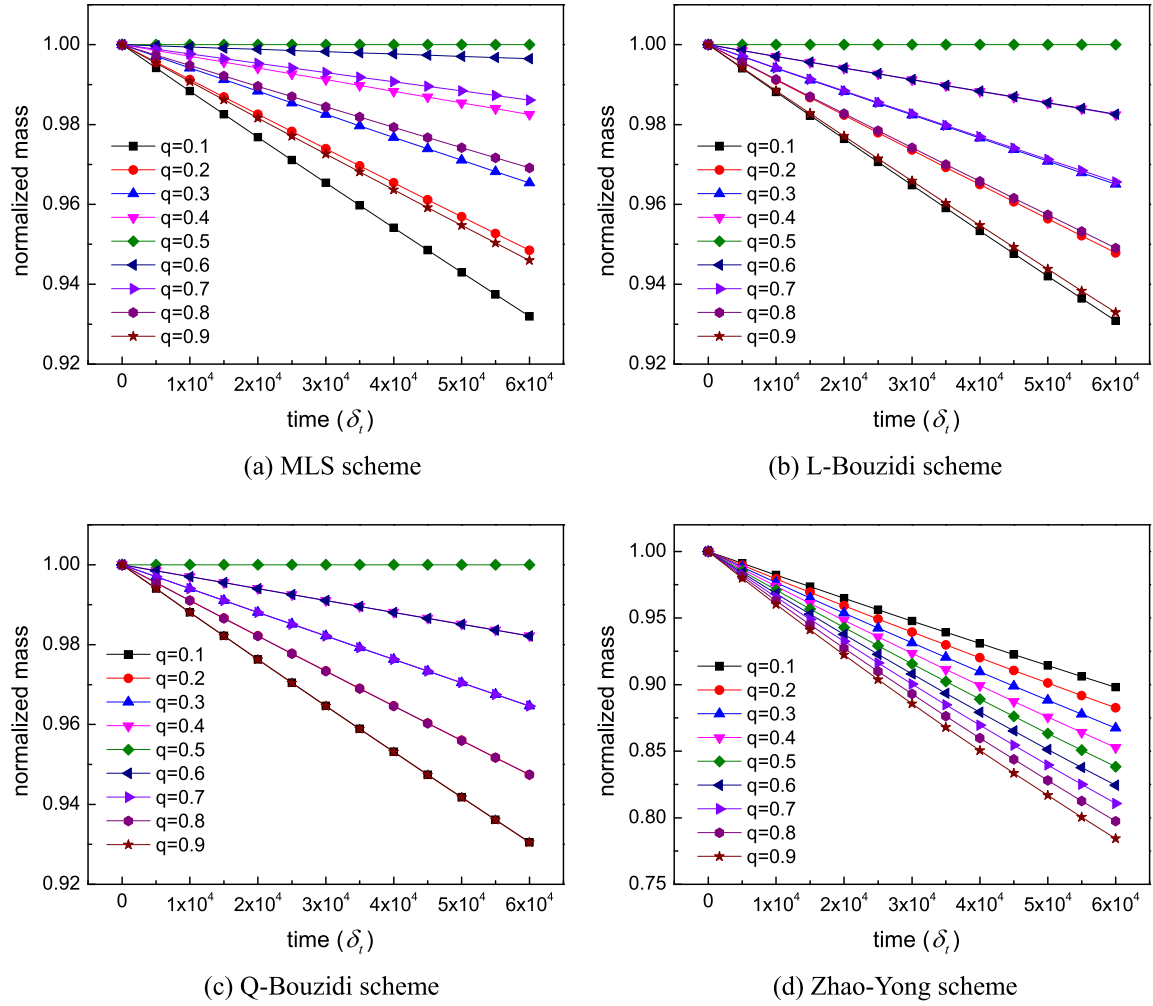


Fig. 4. Simulations of static flow under gravity using different boundary schemes. Variations of the normalized mass of the system against time.

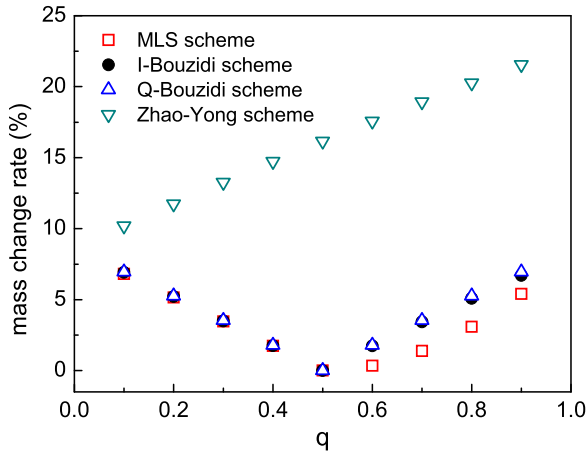


Fig. 5. Simulations of static flow under gravity using different boundary schemes. Variations of the averaged mass change rate against the fraction q defined by Eq. (6).

action force is defined as follows:

$$\mathbf{F} = -G \psi(\mathbf{x}) \sum_{\alpha} w_{\alpha} \psi(\mathbf{x} + \mathbf{e}_{\alpha} \delta_t) \mathbf{e}_{\alpha}, \quad (16)$$

where ψ is the pseudopotential, G is the interaction strength, and w_{α} are the weights. For the nearest-neighbor interactions on the

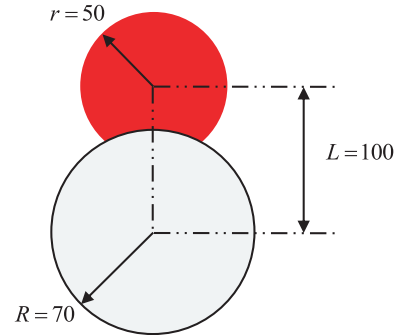


Fig. 6. Initial setting of a droplet resting on a circular cylinder.

D2Q9 lattice, the weights are given by $w_{\alpha} = 1/3$ for $|\mathbf{e}_{\alpha}|^2 = 1$ and $w_{\alpha} = 1/12$ for $|\mathbf{e}_{\alpha}|^2 = 2$.

The pseudopotential is chosen as $\psi(\rho) = \psi_0 \exp(-\rho_0/\rho)$, where ψ_0 and ρ_0 are both set to 1.0, which leads to the liquid density $\rho_L \approx 2.783$ and the vapor density $\rho_V \approx 0.3675$ when the interaction strength is taken as $G = 10/3$ [47]. For this type of pseudopotential, the forcing scheme proposed by Guo *et al.* [44] is employed to incorporate the fluid-fluid interaction force into the LB equation. The contact angle is implemented via a constant virtual wall density [48–50], which does not affect the conclusions in the present work. However, it should be noted that the contact

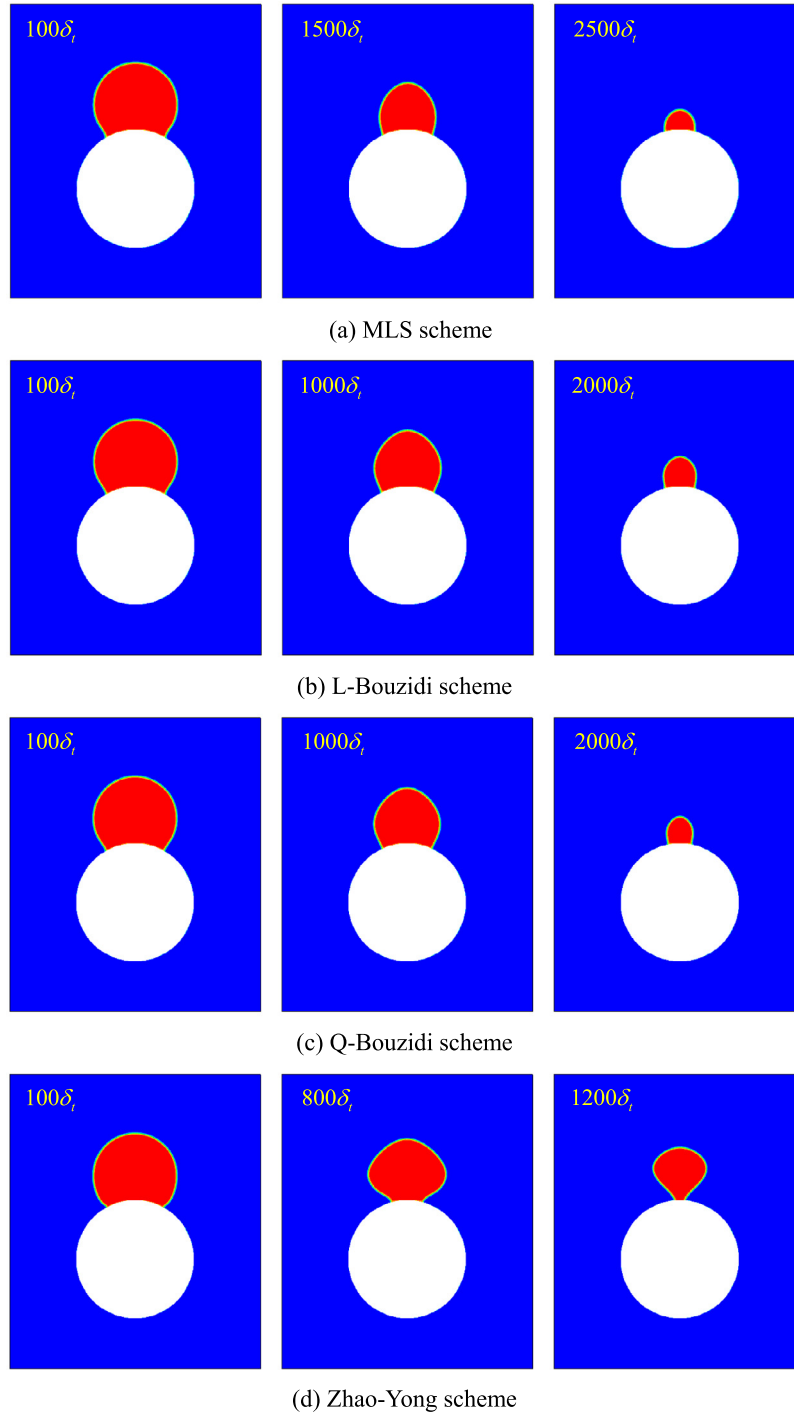


Fig. 7. Snapshots of density contours obtained by different boundary schemes. (a) MLS scheme, (b) L-Bouzidi scheme, (c) Q-Bouzidi scheme, and (d) Zhao-Yong scheme.

angle scheme with a constant virtual wall density usually leads to an unphysical mass-transfer layer near the solid boundary. An improved contact angle scheme, which overcomes this drawback, has been recently proposed by Li *et al.* [51]. In the present simulations, the computational domain is divided into $N_x \times N_y = 300 \times 350$ lattices. A circular cylinder of radius $R = 70$ is located at $(150, 130)$ and a droplet is initially placed on the circular cylinder, as shown in Fig. 6. The periodic boundary condition is utilized in the x and y directions and the kinematic viscosity is taken as $\nu = 0.15$ for both the liquid and vapor phases.

Some snapshots of the density contours obtained by the MLS scheme, the L-Bouzidi scheme, the Q-Bouzidi scheme, and the

Zhao-Yong scheme are displayed in Fig. 7(a), (b), (c), and (d), respectively. From these figures it can be seen that, for all the investigated schemes, the simulated droplet gradually becomes smaller as time goes by, behaving like an evaporating droplet. However, the present simulations are performed using an isothermal pseudopotential LB model and the periodic boundary condition is applied in both the x and y directions, from which it can be concluded that there exists mass leakage at the surface of the cylinder, resulting from the employed curved boundary scheme. Among the investigated boundary schemes, the Zhao-Yong scheme has the advantage of local computations since it is a single-node scheme, while the other schemes usually require the information of two

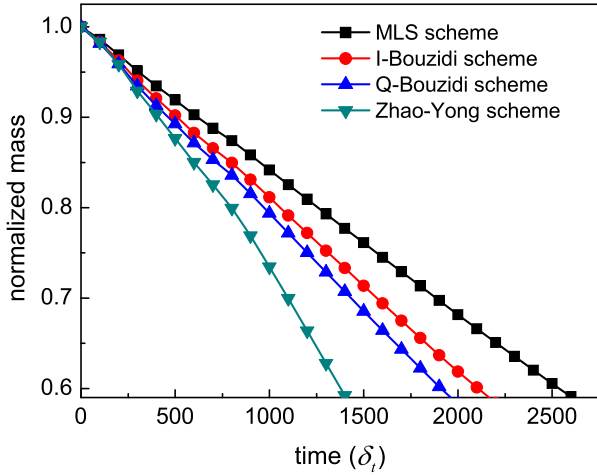


Fig. 8. Simulations of droplet on a circular cylinder using different boundary schemes. Variations of the normalized mass of the system against time.

or more neighboring nodes. Similar to Figs. 4(d), 7(d) also shows that the Zhao-Yong scheme suffers from the violation of mass conservation although it does not involve any interpolation or extrapolation. Particularly, the mass loss rate caused by the Zhao-Yong scheme is much larger than those caused by the other schemes, as shown in Fig. 8, which plots the variations of the normalized mass of the system against time. The normalized mass is defined as $\bar{m} = m/m_0$, where m is the transient mass of the system and m_0 is the initial mass of the system at $t = 0$. From Fig. 8 we can also observe that the MLS scheme performs best among the investigated schemes, i.e., the mass of the system decreases relatively slowly in the test using the MLS scheme. Besides, it can be found that the performance of the Bouzidi scheme is better than that of the Zhao-Yong scheme, but worse than that of the MLS scheme.

By comparing the numerical results of two-phase LB simulations in this section with the results of single-phase LB simulations displayed in Section 3.1, we can find that the mass in two-phase LB simulations decreases significantly faster than that in single-phase LB simulations. Taking the Zhao-Yong scheme as an example, in the present test it yields 40% mass loss after just 1500 time steps, as shown in Fig. 8. Such a rapid decrease of mass in two-phase LB simulations is probably due to the contact between the liquid-vapor interface and the curved solid boundary, which speeds up the mass leakage of the system. When the curved boundary is completely immersed in the liquid or vapor phase, the situation may be similar to that in single-phase LB simulations.

4. Modified mass-conservative schemes

4.1. Formulations of modified curved boundary schemes

In this section, two modified mass-conservative curved boundary schemes are presented. On the basis of the observation that the MLS scheme yields the smallest mass loss among the investigated curved boundary schemes, the modified boundary schemes are given based on the MLS scheme. According to the MLS scheme, the unknown distribution functions at boundary nodes are determined by

$$f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t) = (1 - \chi) f_{\alpha}^*(\mathbf{x}_b, t) + \chi f_{\alpha}^{eq}(\mathbf{x}_s, t), \quad (17)$$

where $f_{\alpha}^{eq}(\mathbf{x}_s, t)$ is defined by Eq. (8) and the parameter χ is given by Eqs. (11) and (10) for $q < 0.5$ and $q \geq 0.5$, respectively. A simple treatment to conserve the mass of a system is to add the mass leakage back to the system. Such an idea was firstly used by Aidun and Lu [52] for suspension flows when employing the

halfway bounce-back scheme for moving boundaries. Similar treatments can also be found in Refs. [10,53]. If the mass leakage is averagely added to the known distribution functions, the momentum at the boundary node may be changed. Hence the mass leakage formulated by Eq. (15) can be added to the rest distribution function and we obtain the following equation:

$$f_0(\mathbf{x}_b, t + \delta_t) = f_0^*(\mathbf{x}_b, t) + \sum_{\text{outgoing}} f_{\alpha}^*(\mathbf{x}_b, t) - \sum_{\text{incoming}} f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t), \quad (18)$$

where $f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t)$ are calculated by Eq. (17). In the remaining of the present paper, this modified MLS scheme is referred to as the modified scheme A. Such a modification is simple, but a possible limitation should also be pointed out here, i.e., the rest distribution function $f_0(\mathbf{x}_b, t + \delta_t)$ may become negative in certain cases.

On the other hand, the mass of a system can also be conserved by guaranteeing the balance between the amount of mass carried by the outgoing and incoming distribution functions at each boundary node. When $\Delta m(\mathbf{x}_b, t + \delta_t)$ in Eq. (15) is set to zero, we can obtain

$$\sum_{\text{incoming}} f_{\bar{\alpha}}(\mathbf{x}_b, t + \delta_t) = \sum_{\text{outgoing}} f_{\alpha}^*(\mathbf{x}_b, t). \quad (19)$$

To satisfy the above requirement, an additional parameter is needed. Inspired by the work of Bao et al. [20], we can redefine the density in the fictitious equilibrium distribution function of ghost nodes

$$f_{\alpha}^{eq}(\mathbf{x}_s, t) = \omega_{\alpha} \rho(\mathbf{x}_s, t) \left[1 + \frac{\mathbf{e}_{\alpha} \cdot \mathbf{u}_s}{c_s^2} + \frac{(\mathbf{e}_{\alpha} \cdot \mathbf{u}_b)^2}{2c_s^4} - \frac{(\mathbf{u}_b \cdot \mathbf{u}_b)}{2c_s^2} \right], \quad (20)$$

where the density has been changed to a fictitious density $\rho(\mathbf{x}_s, t)$. Substituting Eq. (17) into Eq. (19), the requirement given by Eq. (19) can be transformed to

$$\sum_{\text{outgoing}} \chi f_{\alpha}^*(\mathbf{x}_b, t) = \sum_{\text{outgoing}} \chi f_{\alpha}^{eq}(\mathbf{x}_s, t). \quad (21)$$

Combining Eq. (21) with Eq. (20), the fictitious density in Eq. (20) can be obtained

$$\rho(\mathbf{x}_s, t) = \frac{\sum_{\text{outgoing}} \chi f_{\alpha}^*(\mathbf{x}_b, t)}{\sum_{\text{outgoing}} \chi \omega_{\alpha} \left[1 + \frac{\mathbf{e}_{\alpha} \cdot \mathbf{u}_s}{c_s^2} + \frac{(\mathbf{e}_{\alpha} \cdot \mathbf{u}_b)^2}{2c_s^4} - \frac{(\mathbf{u}_b \cdot \mathbf{u}_b)}{2c_s^2} \right]}. \quad (22)$$

Since the parameter χ depends on $\mathbf{e}_{\alpha}$ through the fraction q , it cannot be moved out from the summation sign in Eqs. (21) and (22).

There is an important difference between the above formulation and that of Bao et al. [20]. To obtain the fictitious density, Bao et al. assumed that the outgoing distribution functions, which are known at each boundary node, satisfy Eq. (20) (such an assumption was illustrated above Eq. (12) in Ref. [20]). Then they obtained

$$\sum_{\text{outgoing}} f_{\alpha}^*(\mathbf{x}_b, t) = \sum_{\text{outgoing}} f_{\alpha}^{eq}(\mathbf{x}_s, t). \quad (23)$$

The above equation is just Eq. (12) in Ref. [20]. Contrarily, our formulation given by Eq. (21) is derived from the requirement of Eq. (19), and it has been emphasized that the parameter χ in Eq. (21) is dependent on $\mathbf{e}_{\alpha}$ through the fraction q . Hence χ in Eqs. (21) and (22) cannot be moved out from the summation over α . In the following part, we will show that Eq. (23) still leads to mass leakage in two-phase LB simulations, although this equation seemingly works well for single-phase LB simulations in the study of Bao et al. [20]. The modified MLS scheme based on Eqs. (20) and (22) is referred to as the modified scheme B.

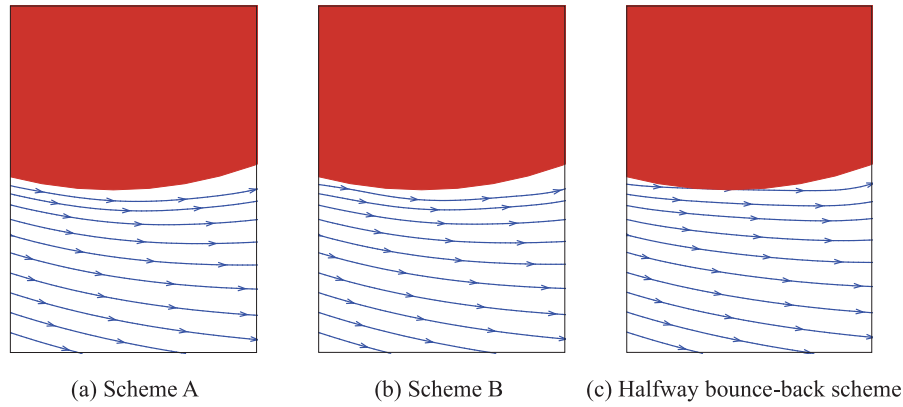


Fig. 9. Comparison of local velocity vectors around the bottom of the circular cylinder at $Re = 40$ between the modified curved boundary schemes and the halfway bounce-back scheme.

Table 1
Comparisons of the wake length, the separation angle, and the drag coefficient.

References	Re = 20			Re = 40		
	$2L/D$	θ_s	C_d	$2L/D$	θ_s	C_d
Scheme A	1.90	41.35	2.130	4.61	53.93	1.590
Scheme B	1.91	41.56	2.133	4.64	53.93	1.593
MLS scheme	1.89	41.15	2.130	4.60	53.85	1.589
Ref. [57]	1.88	43.7	2.045	4.69	53.8	1.522
Ref. [58]	1.90	40.78	2.104	4.64	50.38	1.568
Ref. [59]	1.89	42.95	2.144	4.52	53.86	1.589

4.2. Numerical validation

In this section, numerical simulations are performed to validate the modified curved boundary schemes. Firstly, we employ a single-phase test, i.e., the test of steady flow past a circular cylinder, to examine the numerical accuracy of the modified schemes. The computational domain is chosen as $N_x \times N_y = 40D \times 25D$, where $D = 40$ is the diameter of the circular cylinder located at $(15D, 12.5D)$. The free-stream velocity is taken as $U_0 = 0.1$ and the Reynolds number is defined as $Re = U_0 D / \nu$. The inlet boundary is given by a uniform velocity profile, which is implemented by the Zou-He boundary scheme [54], while the fully developed boundary condition is employed at the outlet boundary. The periodic boundary condition is applied to the upper and lower boundaries and the modified curved boundary schemes are used at the surface of the circular cylinder.

The non-dimensional length of the recirculation region, the separation angle, and the drag coefficient obtained by the modified schemes A and B are compared with some published results in Table 1. The non-dimensional length of the recirculation region is defined as $2L/D$, where L is measured from the rearmost point of the cylinder. The drag coefficient is defined as $C_d = \bar{F}_x / 0.5 \rho U_0^2 D$, where $\bar{F}_x$ is the x -component of the force $\bar{\mathbf{F}}$ exerted on the cylinder, which is calculated using the momentum-exchange method [55,56]. From the table we can see that the numerical results obtained by the modified schemes A and B are both in good agreement with those reported in the previous studies.

Moreover, to illustrate the differences between the modified MLS schemes and the halfway bounce-back scheme, Fig. 9 presents a comparison of the local velocity vectors around the bottom of the circular cylinder at $Re = 40$ between the results of the modified schemes and those of the halfway bounce-back scheme. As shown in the figure, the halfway bounce-back scheme leads to unphysical results below the circular cylinder as the velocity obtained by

the staircase approximation penetrates the real cylindrical surface, which there is no such a phenomenon in the numerical results of the modified schemes A and B.

Now we turn our attention to the capability of the modified curved boundary schemes in conserving mass in two-phase LB simulations. The test of a droplet placed on a circular cylinder is employed and the initial setting can be found in Fig. 6. Three cases with different contact angles are considered, i.e., $\theta = 30^\circ, 90^\circ$, and 120° . The grid system and the simulation parameters are the same as those used in Section 3.2. Some snapshots of the density contours obtained by the modified scheme A for the three cases are displayed in Fig. 10. By comparing the numerical results in Fig. 10 and those in Fig. 7, we can see that the droplet can achieve a stable state when the modified scheme A is employed, rather than behaving like an evaporating droplet when the original MLS scheme is used, which indicates that the modification formulated by Eq. (18) has eliminated the mass leakage at the surface of the circular cylinder.

Fig. 11 displays the variations of the normalized mass of the system during the simulations using the modified schemes A and B, respectively. For both modified schemes, the maximum variation of the normalized mass of the system is very small (about 0.01%) and it appears at the early stage of simulation, which is expected because the liquid-vapor interface is initially a sharp interface and it develops into a diffuse interface with a finite thickness during the early stage of simulation. Such a change causes a small fluctuation of the mass of the system. After the early stage, the variations of the normalized mass of the system are much smaller, as can be seen in Fig. 11. In addition, we can also observe some slight differences between the results of different cases ($\theta = 30^\circ, 90^\circ$, and 120°), which are attributed to the fact that the perimeter/length of the liquid-vapor interface is slightly different in these cases. To sum up, it is numerically demonstrated that both the modified scheme A and scheme B are capable of conserving mass in two-phase LB simulations, although they are devised based on the MLS boundary scheme from different points of view.

A comparison of the modified scheme B and the scheme of Bao et al. [20] is made in Fig. 12 by taking the case of $\theta = 90^\circ$ as an example. From the figure it can be seen that the scheme of Bao et al. still results in mass leakage in two-phase LB simulations although it performs much better than the original MLS scheme (cf. Fig. 7a). The reason for such a phenomenon has been previously mentioned, i.e., the fictitious density in the scheme of Bao et al. [20] was obtained by assuming that the outgoing distribution functions satisfy Eq. (20). Such an assumption leads to the aforementioned Eq. (23), which is equivalent to moving the parameter χ (dependant on $\mathbf{e}_\alpha$) in Eq. (21) out from the summation over α . Ac-

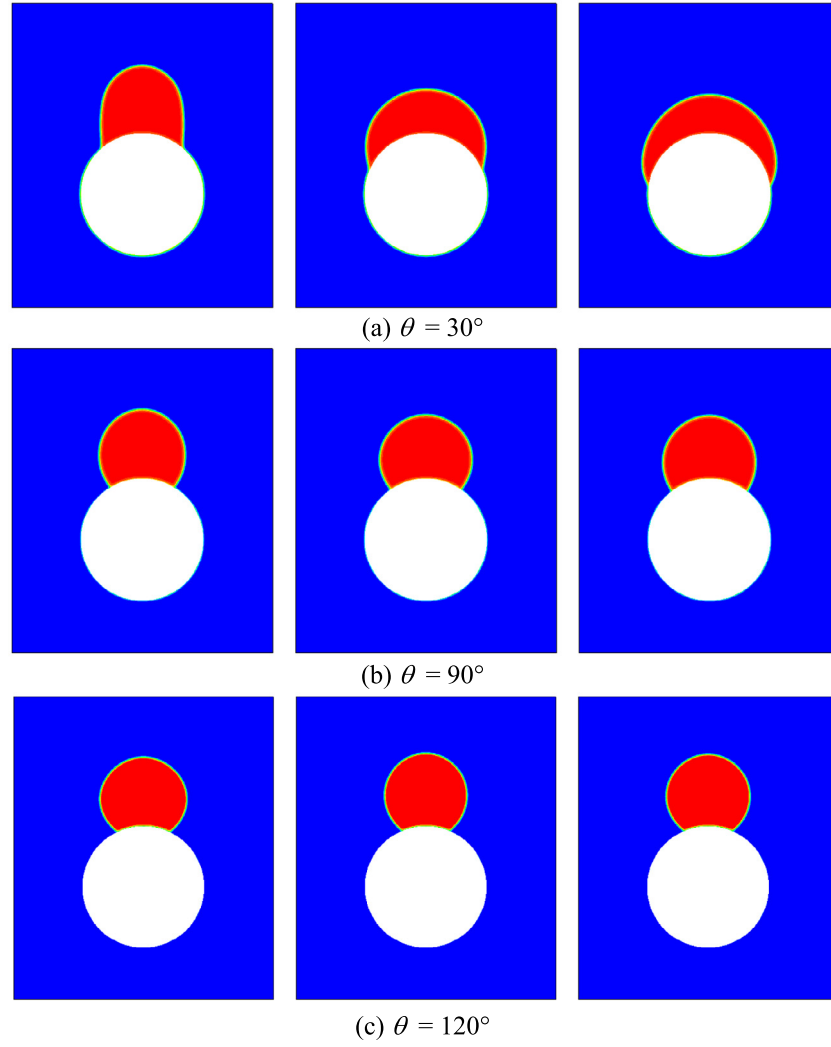


Fig. 10. Simulations of droplet on a circular cylinder using the modified curved boundary scheme A. Snapshots of density contours under the static contact angles $\theta = 30^\circ$, 90° , and 120° , respectively. From left to right: $t = 1000\delta_t$, $5000\delta_t$, $20000\delta_t$.

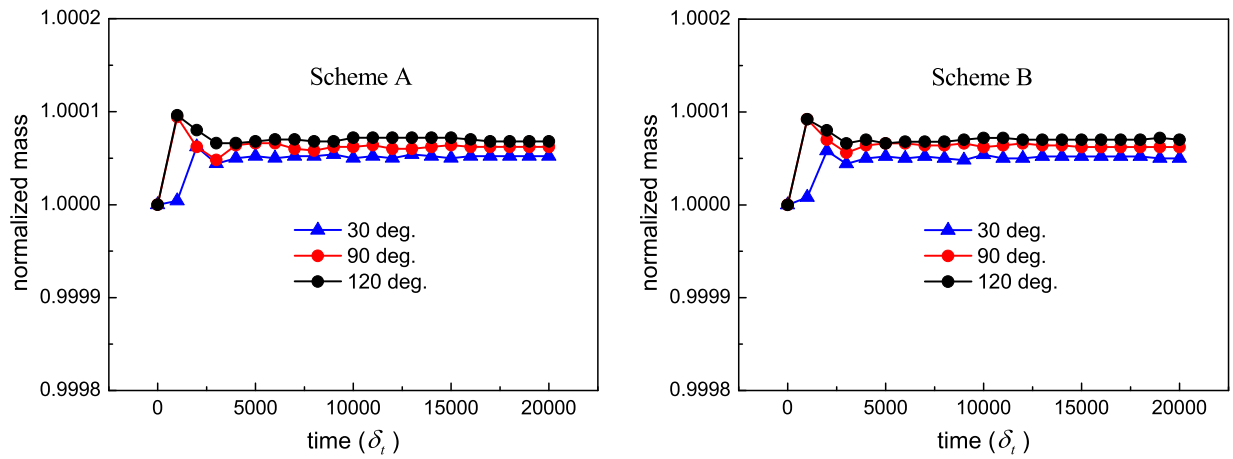


Fig. 11. Variations of the normalized mass of the system under the static contact angles $\theta = 30^\circ$, 90° , and 120° , respectively.

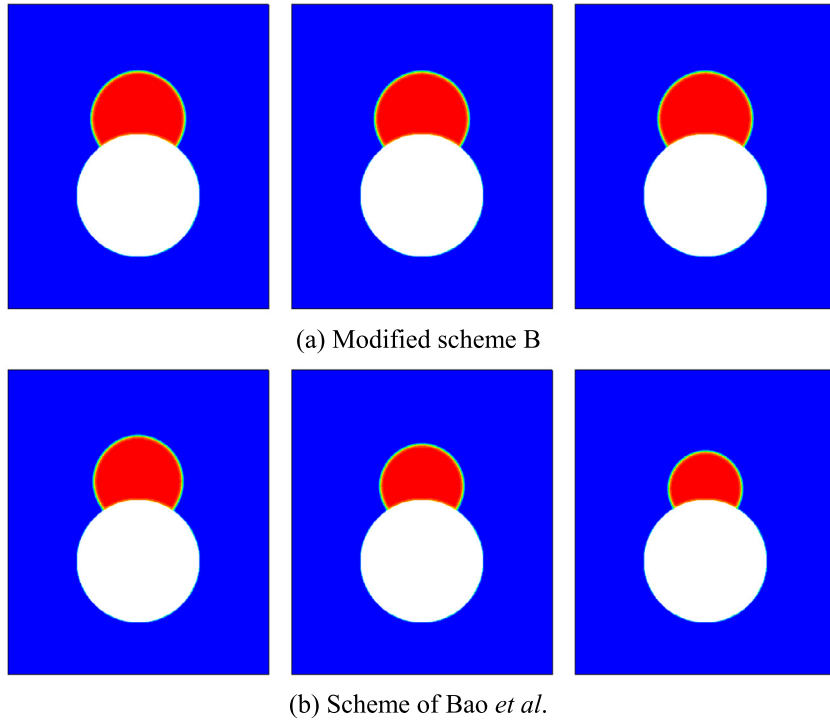


Fig. 12. A comparison of the results obtained by the modified curved boundary scheme B and the scheme of Bao *et al.* [20]. From left to right: $t = 10000\delta_t$, $5 \times 10^4\delta_t$, $10^5\delta_t$. The static contact angle is $\theta = 90^\circ$.

cordingly, it still suffers from the violation of mass conservation, although such a defect was not detected by the single-phase LB simulations in Ref. [20].

5. Summary

In this paper, we have investigated the performances of three well-known types of curved boundary schemes in two-phase LB simulations. Through modeling a droplet resting on a circular cylinder, it has been found that, for all the investigated schemes, the simulated droplet rapidly “evaporates” under the non-slip and isothermal condition, revealing that there exists serious mass leakage at the cylindrical surface due to the curved boundary schemes. The numerical results showed that the MLS boundary scheme yields the smallest mass loss among the investigated curved boundary schemes, while the mass loss caused by the Zhao-Yong scheme, which is a single-node boundary scheme and has the advantage of local computations, is much larger than those of other schemes.

On the basis of the numerical investigation, we have formulated two modified mass-conservative boundary schemes for curved boundaries based on the MLS boundary scheme. The modified scheme A employs a modification to the rest distribution function of the boundary node so as to add the mass leakage back to the system, while the modified scheme B is devised by guaranteeing the balance between the amount of mass carried by the outgoing and incoming distribution functions at each boundary node. Numerical simulations have been carried out to validate the modified curved boundary schemes. It has been shown that the two modified schemes exhibit better performance than the halfway bounce-back scheme according to the modeling of steady flow past a circular cylinder. Moreover, the simulations of a droplet resting on a circular cylinder demonstrate that the two modified schemes are capable of conserving mass in two-phase LB simulations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Y. Yu: Methodology, Software, Writing - original draft. **Q. Li:** Conceptualization, Supervision, Writing - review & editing. **Z.X. Wen:** Writing - review & editing.

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Appendix

In this appendix, we present some results of combining the Bouzidi curved boundary scheme with the modification given by Eq. (18). The simulation setup and parameters are the same as those described in Fig. 6. The L-Bouzidi scheme [Eq. (12)] is combined with the modification Eq. (18) and the case of contact angle $\theta = 90^\circ$ is studied as an example. Fig. 13 displays the snapshots of density contours obtained by the modified L-Bouzidi scheme at $t = 1000\delta_t$, $5000\delta_t$, and $20000\delta_t$, respectively. It can be seen that the droplet can achieve a stable state when the modification Eq. (18) is applied to the L-Bouzidi scheme. Quantitatively, the variation of the normalized mass of the system against time is plotted in Fig. 14, which shows that during the simulation the maximum variation of the normalized mass of the system is about 0.01% and demonstrates that the modified L-Bouzidi scheme is also able to conserve mass in two-phase LB simulations.

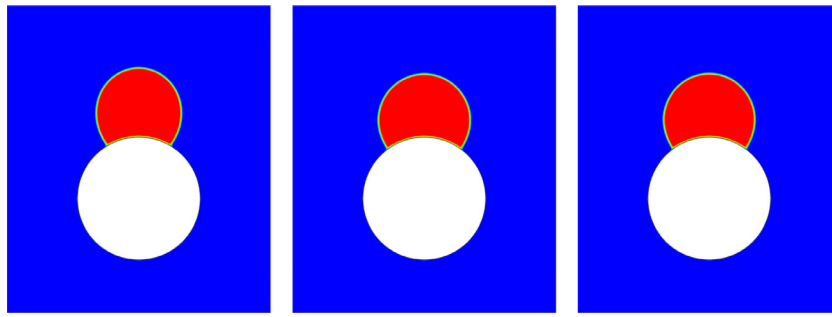


Fig. 13. Simulation of droplet on a circular cylinder using the L-Bouzidi scheme with the modification Eq. (18) under the static contact angles $\theta = 90^\circ$. From left to right: $t = 1000\delta_t$, $5000\delta_t$, $20000\delta_t$.

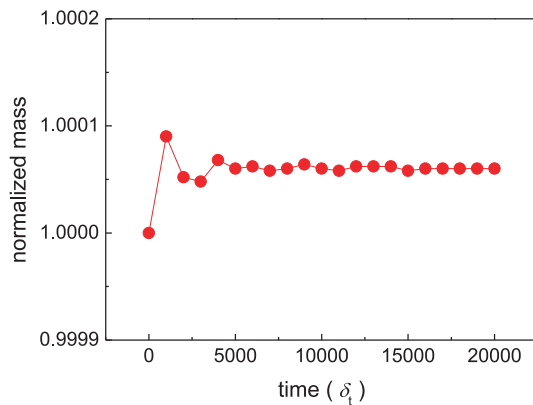


Fig. 14. Variation of the normalized mass of the system under the static contact angles $\theta = 90^\circ$. The L-Bouzidi scheme with the modification Eq. (18) is applied.

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